Queueing Theory 'OR' Waiting Line Theory

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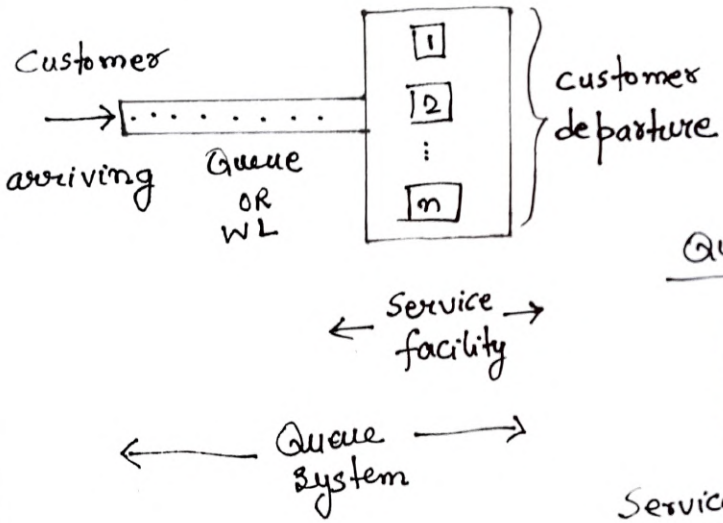
Introduction : A flow of customer from infinite / finite population towards the service facility forms a queue (waiting Line).

Why Queue Form :

(i) When no. of customers exceed no. of server.

(ii) Server do not work effieiently and take more time than prescribed to serve a customer.

Queue Model :



Customer : The arriving unit that requires some services to be performed is called customer. The customer may be person, machines, vehicles etc.

Queue : stands for the number of customers waiting to be serviced. This does not include the customer being serviced.

Service facility : The process or system that provides services to the customers is termed as service channel or service facility.

Queue System :

A queue system can be completly described by :

(1) The Input (Arrival pattern)
(2) The service mechanism (service pattern)
(3) The Queue discipline
(4) Customer's behaviour.

(i) **The Input (Arrival pattern)** : Input describe the way in which the customers arrive and join the system. Generally, customer arrive in a more or less random fashion, which is not possible to predict. Thus, the arrival pattern can be described in terms of probabilities, and consequently, the probability distribution for inter-arrival times (the time between two successive arrivals) must be defined. We deal with those queueing system in which the customers arrive in poisson fashion.

(ii) Service Mechanism : This means, the arrangement of service facility to serve customers.

If there is an infinite no. of servers, then all the customers are served instantaneously on arrival, and there will be no queue.

If no. of servers is finite then the customers are served according to spefic order, with service time as a constant or random variable.

(iii) Queue discipline : It is a rule according to which the customers are selected for service when a queue has been formed. The most common discipline are :

a) First come First serve (FCFS) or FIFO
(b) LCFS or LIFO
(c) SIRO (service in random order)
(d) Priority.

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(iv) Customer's Behaviour : The customers generally behave in the following ways.

(a) Patience

(b) Impatience

(i) Balking

(ii) Reneging

(iii) Jockeying

Mean Arrival Rate (λ): No. of customer arriving per unit of time.

Mean service rate (μ): No. of customer served per unit of time.

Operating Characteristics of Queue System

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(1) Queue Length (L_q) : The average no. of customer in the queue waiting to get service. This excludes the customers being served.

(2) System Length (L_s) : The average no. of customers in the system including those waiting as well as those being served.

(3) Waiting time in the queue (W_q) : The average time for which a customer has to wait in the queue to get service.

(4) Waiting time in system (W_s) : The average total time spent by a customer in the system from the moment he arrives till he leave the system. It is taken to be the waiting time plus service time.

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Traffic Intensity (or utilization factor) : denoted by ρ.

It is the proportion of time a server actually spends with the customers.

It is the ratio of mean arrival rate and mean service rate.

$$\text{Traffic intensity } (\rho) = \frac{\text{Mean arrival rate}}{\text{Mean service rate}}$$

$$\boxed{\rho = \frac{\lambda}{\mu}}$$

The unit of traffic intensity is Erlang.

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Transient state and steady state : A system is said to be in a transient state when its operating characteristics are depending on time.

A steady state system is the one in which the behaviour of the system is independent of time. Let $P_n(t)$ denote the probability that there are n customers in the system, at time t. Then in steady state,

$$\lim_{t \to \infty} p_n(t) = p_n \quad \text{(independent of } t\text{)}$$

$$\Rightarrow \quad \frac{d p_n(t)}{dt} = \frac{d p_n}{dt}$$

$$\Rightarrow \quad \lim_{t \to \infty} p_n'(t) = 0$$

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Kendall's Notation for Representing Queue Models

Generally, queueing model may be completely specified in the following symbol form (a/b/c):(d/e) where,

a = probability law for the arrival (inter-arrival) time. (arrival distribution)

b = probability law according to which the customers are being served. (service time distribution)

c = number of channels (or service stations)

d = Capacity of system i.e. max. no. of customer allowed in the system (in service and waiting)

e = queue discipline (FCFS, LIFO)

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Symbol Used for a & b :

M → Markovian (Poisson) arrival or departure distribution or exponential inter arrival or service time distribution.

E_K → Erlangian or gamma interval or service time distribution with parameter K.

GI → General independent arrival distribution.

D → Deterministic interval or service time

G → General departure distribution.

Symbol Used for C :

1 → Single server

* S → Fixed no. of server.

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Symbol Used for d :

∞ → Infinite no. of customer

N → finite no. of customer.

Symbol Used for e :

FCFS → First come first serve

LCFS → Last come first serve

SIRO → service in random order

GD → General discipline

Classification of Queueing Models.

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The queueing models are classified as follows:

Model I : (M/M/1) : (∞/FCFS) : This denotes Poisson arrival, poisson departure, single server, Infinite capacity and First come first serve discipline.

Model II : (M/M/1) : (N/FCFS) : This denotes poisson arrival, poisson departure, single server, Capacity of system is limited (finite), FCFS discipline.

Model III : Multiservice Model (M/M/S) : (∞/FCFS) : This model takes the no. of service channel as S.

Model IV : (M/M/S) : (N/FCFS) : This model is same as model III, except the max. of customer in the system is limited to N, (N > S)

Model I : (M/M/1) (∞/FCFS) or Birth and Death Model

λ = Mean arrival rate (no. of arrival per unit time)

μ = Mean service rate (no. of customer served per unit time)

(i) Traffic intensity, $\boxed{\rho = \frac{\lambda}{\mu}}$

(ii) Expected (average) no. of units in the system L_s

$$\boxed{L_s = \frac{\frac{\lambda}{\mu}}{1-\frac{\lambda}{\mu}} = \frac{\lambda}{\mu-\lambda}}$$

(iii) Expected (average) queue length L_q.

$$\boxed{L_q = L_s - \frac{\lambda}{\mu} = \frac{\lambda^2}{\mu(\mu-\lambda)}}$$

(iv) Expected Waiting time in the queue, W_q

$$\boxed{W_q = \frac{\lambda}{\mu(\mu-\lambda)}}$$

(v) Expected Waiting line in the system, W_s

$$W_s = W_q + \frac{1}{\mu} = \frac{1}{\mu - \lambda}$$

(vi) Expected waiting time of a customer who has to wait $(W/W>0)$

$$(W/W>0) = \frac{1}{\mu-\lambda} = \frac{1}{\mu(1-\rho)}$$

(vii) Expected length of non-empty queue,

$$(L/L>0) = \frac{\mu}{\mu-\lambda} = \frac{1}{1-\rho}$$

(viii) Probability of Waiting time in the queue $\geq t$

$$= \int_t^\infty \lambda(1-\lambda) e^{-(\mu-\lambda)w} dw$$

(ix) Probability of queue size $\geq N = \rho^N$

$$= \int_t^\infty \rho(\mu-\lambda) e^{-(\mu-\lambda)w} dw$$

(x) Probability (Waiting time in the system $\geq t$)

$$= \int_t^{\infty} \rho(\mu-\lambda) e^{-(\mu-\lambda)w} \, dw$$

Littles formulas

$$L_s = \lambda W_s$$

$$W_q = W_s - \frac{1}{\mu}$$

$$L_q = L_s - \frac{\lambda}{\mu}$$

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Q: A T.V. mechanic finds that the time spent on his Jobs has an exponential distribution with mean 30 minutes, if he repairs sets in the order in which they come in. If the arrival of sets is approximately poisson with an average rate of 10 per eight-hour day, what is the mechanics expected idle time each day? How many Jobs are ahead of the average sets just brought in?

Solution: $\mu = \frac{1}{30}$, $\lambda = \frac{10}{8 \times 60} = \frac{1}{48}$. $\Rightarrow \frac{\lambda}{\mu} = \frac{30}{48}$

Expected no. of Jobs are,

$$L_s = \frac{\frac{\lambda}{\mu}}{1 - \frac{\lambda}{\mu}} = \frac{\lambda}{\mu - \lambda} = \frac{\frac{1}{48}}{\frac{1}{30} - \frac{1}{48}}$$

$$= \frac{\frac{1}{48}}{\frac{48-30}{30 \times 48}} = \frac{1}{\cancel{48}} \times \frac{\overset{5}{\cancel{30} \times \cancel{48}}}{\cancel{48}_3}$$

$L_s = \frac{5}{3}$

$L_s = 1\frac{2}{3}$ Jobs.

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Since the fraction of the time the mechanic is busy $= \frac{\lambda}{\mu}$

∴ The no. of hours for which the repairman remains busy in an eight-hour day.

$= 8\left(\frac{\lambda}{\mu}\right) = 8 \times \frac{30}{48} = 5$ hours.

Therefore, the time for which the mechanic remain idle in an eight-hour day $= (8 - 5)$ hours.

$= 3$ hours.

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Model-I (M/M/1) : (∞/FCFS) Numerical Problem-2

Q: Arrivals at a telephone booth are considered to Poisson, with an average time of 10 minutes between one arrival and the next. The length of a phone call is assumed to be distributed exponentially with mean three minutes.

(i) What is the average length of the queue that forms from time to time ?

(ii) The telephone department will install a second booth when convinced that an arrival would have to wait at least three minutes for the phone. By how much time must the flow of arrivals be increased in order to justify a second booth ?

(iii) Estimate the fraction of a day that the phone will be in use.

(iv) Find the average number of units in the system.

solution :

(i) $\lambda = \frac{1}{10}$, $\mu = \frac{1}{3}$

Average length of non-empty queue,

$$(L/L>0) = \frac{\mu}{\mu-\lambda} = \frac{\frac{1}{3}}{\frac{1}{3}-\frac{1}{10}} = \frac{\frac{1}{3}}{\frac{7}{3\times 10}} = \frac{10}{7} = 1.43 \text{ person}$$

(ii) $W_q = 3$, $\mu = \frac{1}{3}$, $\lambda = \lambda'$

$$W_q = \frac{\lambda}{\mu(\mu-\lambda)}$$

$$\Rightarrow 3 = \frac{\lambda'}{\frac{1}{3}\left(\frac{1}{3}-\lambda'\right)}$$

$$\Rightarrow \frac{1}{3} - \lambda' = \lambda'$$

$$\Rightarrow \frac{1}{3} = 2\lambda'$$

$$\Rightarrow \lambda' = \frac{1}{6} = 0.16$$

Hence, increase in the arrival rate $= 0.16 - 0.10$

$= 0.06$ arrival per min.

(iii) The fraction of a day that the phone will be in busy = Traffic Intensity.

$$\rho = \frac{\lambda}{\mu} = \frac{\frac{1}{10}}{\frac{1}{3}} = 0.3.$$

(iv) Average no. of unit in the system,

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{\frac{1}{10}}{\frac{1}{3} - \frac{1}{10}} = \frac{1}{10} \times \frac{10 \times 3}{7} = 0.43 \text{ person.}$$